

* Machine Learning Workflow:

1. Problem Statement
2. Data Collection (file base, data base)
3. Data cleaning / Transformation / preprocessing.

- Data types
- Null types
- Duplicated values
- column names

4. Feature Engineering

- adding a column
- updating a column
- deleting a column
- Converting categorical to numerical.

5. Model Validation (only for Supervised learning).

6. Model selection.

7. Training the Model

8. Model Evaluation

9. Deployment & Monitoring.

Machine Learning Algorithms:-

Algorithm 1:-

Linear Regression:- When our prediction point / value is continuous numerical in that case we have

to use regression Algorithms.

Ex:-

x_1	x_2	x_3	x_4	x_5	x_6
size	Type of House	Area	Amenities	parking	Type of building

Rent $\rightarrow y$ Continuous numerical.

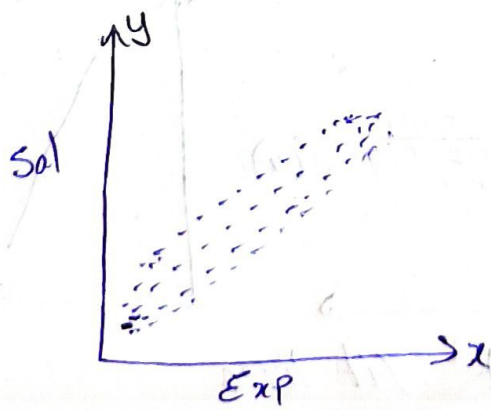
Independent Variables :- x (features) input
Dependent Variables :- y (target) output.

Note:- Always y should be dependent on x , x should not depend on each other.

$\rightarrow$ If we want to use simple linear regression the data has to pass some assumptions.

1. Always x & y should be colinear to each other. It may be +ve colinear or -ve colinear but not non-colinear.

2. The data contains only one feature & one target.



Exp	Sal
2	30,000
4	75,000
3.5	68,000
3	25,000

What is math behind simple linear regression?

It is linear line equation

$$y = mx + c$$

Where, $m = \text{slope}$
 $c = \text{constant}$

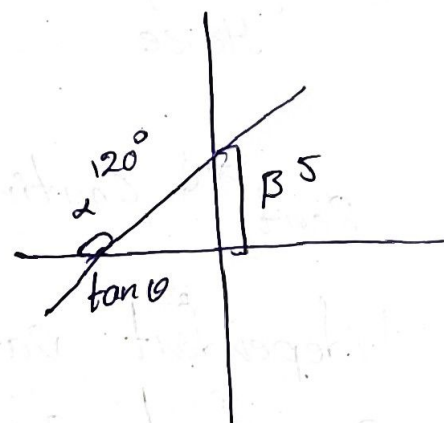
SLR Equation is

$$\hat{y} = \alpha x + \beta$$

Where, α - Co-efficient

β - Intercept

$\alpha = \tan \theta$



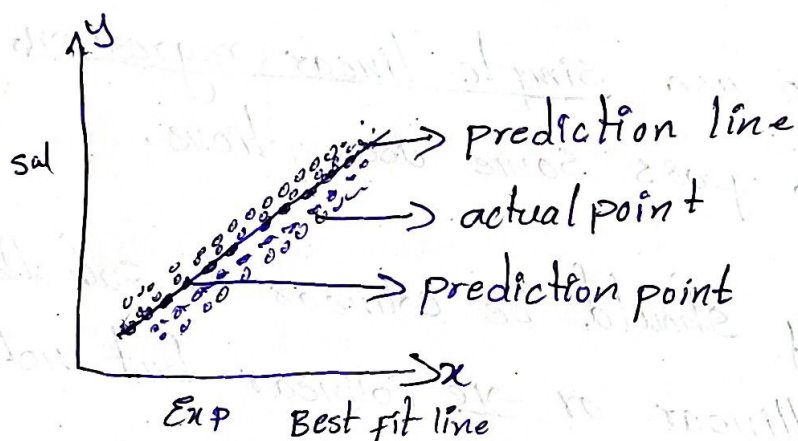
Ex:-

$$\hat{y} = \alpha x + \beta$$

$$\hat{y} = \tan 120^\circ + 5$$

$$\hat{y} = -\sqrt{3}x + 5$$

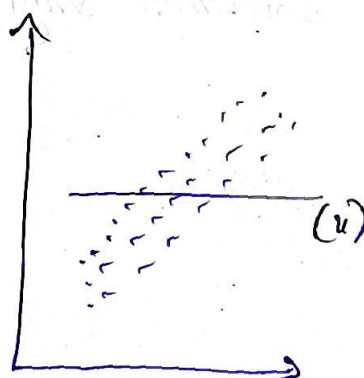
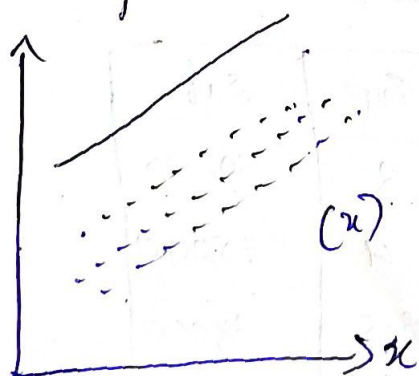
$$\hat{y} = -1.73x + 5$$



blue dots $\rightarrow$ actual points

black dots $\Rightarrow$ prediction points

The prediction line always should be near to the actual point.



No Best fit lines

$$\boxed{\hat{y} = \alpha x + \beta}$$

in ml.

$\alpha \rightarrow$ Co-efficient

$\beta \rightarrow$ Intercept

$$\hat{y} = \tan 120^\circ x + 5$$

$$\hat{y} = -\sqrt{3}x + 5$$

$$\boxed{\hat{y} = -1.773x + 5}$$

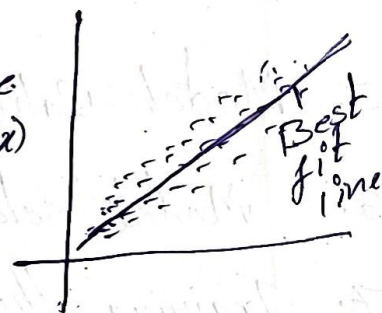
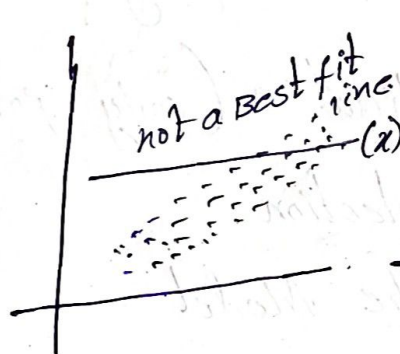
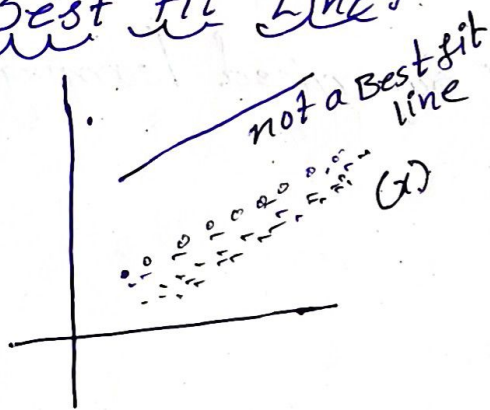
$$\hat{y} = (-1.773 \times 3) + 5$$

$$\boxed{\hat{y} = -0.319}$$

$\pi = 3.142$
Constant.

$\rightarrow \alpha$ and β are hyperparameters, mean the values can be change to improve the model.

Best Fit Line:



The avg distance b/w actual point and predicted line it should be less / minimum.

$\rightarrow$ Always predicted line is dependant on α & β values. so α & β are called as hyper parameters.

hyper-parameters $\rightarrow$ non constant values unlike π which is 3.14 all the time.